

EXPERIMENT–1

IoE Sensor Data Acquisition and Preprocessing using Python in Google Colab

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AIM:

To acquire IoE sensor data, perform preprocessing techniques such as handling missing values, removing duplicates, detecting outliers, normalizing data, and preparing the dataset for AI/ML applications using Python in Google Colab.

OBJECTIVES

- Understand IoE datasets.
- Import data.
- Handle missing values.
- Remove duplicates.
- Detect outliers.
- Normalize data.
- Visualize data.
- Prepare for ML.

THEORY

The Internet of Everything (IoE) integrates people, processes, data and things. Sensor devices continuously generate data such as temperature, humidity, pressure, CO₂ concentration and soil moisture. Raw data may contain missing values, duplicate records and abnormal readings. Data preprocessing improves the quality, consistency and reliability of the dataset before applying Artificial Intelligence and Machine Learning algorithms

Software Requirements

- Google Colab
- Python 3
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-learn

Algorithm

1. Start Google Colab.
2. Import libraries.
3. Upload dataset.
4. Load CSV file.
5. Display dataset.
6. Check and fill missing values.
7. Remove duplicate records.
8. Detect outliers.
9. Normalize numeric values.
10. Visualize data.
11. Save cleaned dataset.

Python Program

Step 1: Import Required Libraries

```
import pandas as pd
```

```
import numpy as np
```

```
import matplotlib.pyplot as plt
```

```
import seaborn as sns
```

```
from sklearn.preprocessing import MinMaxScaler
```

Step 2: Upload Dataset

```
from google.colab import files
```

```
print("Upload sensor_data.csv")
uploaded = files.upload()
# Step 3: Load Dataset
df = pd.read_csv("sensor_data.csv")
# Step 4: Display Dataset
print("\nFirst 5 Records")
print(df.head())
print("\nDataset Information")
print(df.info())
print("\nDataset Shape")
print(df.shape)
# Step 5: Check Missing Values
print("\nMissing Values")
print(df.isnull().sum())
# Step 6: Fill Missing Values with Mean
df.fillna(df.mean(numeric_only=True), inplace=True)
# Step 7: Remove Duplicate Records
duplicates = df.duplicated().sum()
print("\nDuplicate Records:", duplicates)
df.drop_duplicates(inplace=True)
# Step 8: Detect Outliers using IQR
print("\nChecking Outliers")
numeric_cols = df.select_dtypes(include=np.number).columns
for col in numeric_cols:
    Q1 = df[col].quantile(0.25)
    Q3 = df[col].quantile(0.75)
```

$IQR = Q3 - Q1$

$lower = Q1 - 1.5 * IQR$

$upper = Q3 + 1.5 * IQR$

`outliers = df[(df[col] < lower) | (df[col] > upper)]`

`print(f'{col}: {len(outliers)} Outliers')`

Step 9: Normalize Numeric Data

`scaler = MinMaxScaler()`

`df[numeric_cols] = scaler.fit_transform(df[numeric_cols])`

`print("\nNormalized Dataset")`

`print(df.head())`

Step 10: Plot Temperature Graph

`plt.figure(figsize=(8,5))`

`plt.plot(df['Temperature'], marker='o')`

`plt.title("Normalized Temperature Sensor Readings")`

`plt.xlabel("Sample Number")`

`plt.ylabel("Temperature")`

`plt.grid(True)`

`plt.show()`

Step 11: Box Plot

`plt.figure(figsize=(8,5))`

`sns.boxplot(data=df[numeric_cols])`

`plt.title("Boxplot of Sensor Data")`

`plt.xticks(rotation=45)`

`plt.show()`

Step 12: Histogram

`df[numeric_cols].hist(figsize=(12,8))`

```

plt.suptitle("Histogram of Sensor Values")

plt.show()

# Step 13: Correlation Heatmap

plt.figure(figsize=(8,6))

sns.heatmap(df.corr(numeric_only=True),

            annot=True,

            cmap='coolwarm',

            linewidths=0.5)

plt.title("Correlation Heatmap")

plt.show()

# Step 14: Save Cleaned Dataset

df.to_csv("clean_sensor_data.csv", index=False)

print("\nCleaned dataset saved as clean_sensor_data.csv")

# Step 15: Download Cleaned Dataset

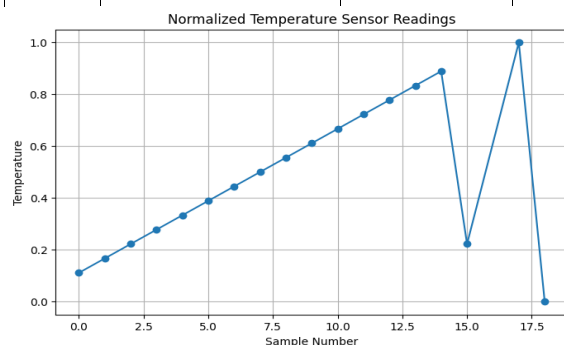
files.download("clean_sensor_data.csv")

print("\nExperiment Completed Successfully!")

```

OUTPUT :

S.no	Temperature	Humidity	pressure	Co2	LightIntensity	SoilMoisture
1	0.11111	0.17777	0.90000	0.058824	0.083333	0.090909
2	0.166666	0.2222	0.8500000	0.078431	0.1333	0.14545
3	0.22222	0.266666	0.80000	0.098039	0.1666	0.1818
4	0.27777	0.3333	0.75	0.117647	0.200	0.21818
5	0.33333	0.3777	0.7000	0.1372549	0.25	0.2727



RESULT:

The IoE sensor dataset was successfully preprocessed and prepared for AI/ML applications.